

Lab 03 - NAND Only Logic

In this lab, you've learned how to convert arbitrary logical equations into NAND only circuits, and why that might be a good thing.

Rubric

Item	Description	Value
Summary Answers	Your writings about what you learned in this lab.	25%
Question 1	Your answers to the question	25%
Question 2	Your answers to the question	25%
Question 3	Your answers to the question	25%

Lab Summary

We learned how to use a breadboard using wires and a chip to build a NAND circuit to power LED lights. We also learned how to apply it in a diagram before building the circuit.

Lab Questions

1. $\sim(A \& B) = \sim A \mid \sim B$ (Eqn 1)

$\sim(A \mid B) = \sim A \& \sim B$ (Eqn 2)

A	B	C	D	$(A \& \sim B) + (C \& D)$	$\sim(\sim(A \cdot \sim(B \cdot B) \cdot \sim(C \cdot D)))$
1	1	1	1	0	0
1	1	1	0	0	0
1	1	0	1	0	0
1	1	0	0	0	0
1	0	1	1	1	1
1	0	1	0	1	1
1	0	0	1	1	1
1	0	0	0	1	1
0	1	1	1	0	0
0	1	1	0	0	0
0	1	0	1	0	0
0	1	0	0	0	0
0	0	1	1	0	0
0	0	1	0	0	0
0	0	0	1	0	0
0	0	0	0	0	0

2. Universalizes the circuit into a single standard type of gate to make it run more efficiently.
3. The breadboard is a more broken down way on how a circuit works or is built. Where as a FPGA is a more complex circuit that has already been pre-built.

Code Submission

Upload a .zip of all your code or a public repository on GitHub.